The Infinite Angle Theorem

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Let the dimensions of a rectangle be l and b where "$b = lx$". Now, look at the figure 1 for better understanding and perspective.

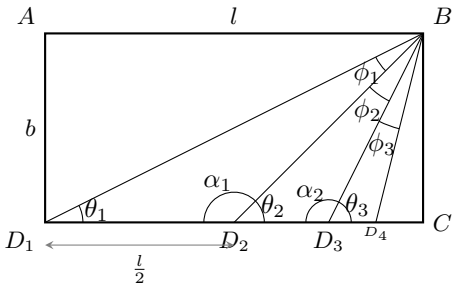


Figure 1: Not to scale

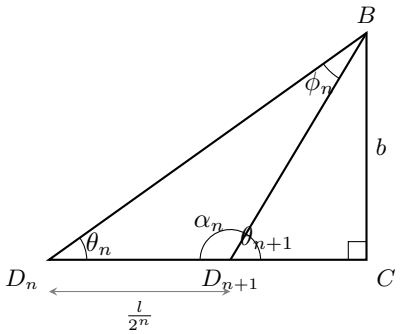


Figure 2: Not to scale

Now, let's take the perspective of nth triangle. Look at Figure 2.
From Figure 2,

$$\operatorname{Tan}\theta_n = \frac{b}{l/2^{n-1}} = \frac{b}{l}(2^{n-1})$$

As $b = lx$,

$$\Rightarrow x = \frac{b}{l}$$

Now, substituting this into $\operatorname{Tan}\theta_n = \frac{b}{l}(2^{n-1})$ gives:

$$\operatorname{Tan}\theta_n = x \cdot 2^{n-1}$$

$$\Rightarrow \theta_n = \operatorname{Tan}^{-1}\left[x \cdot 2^{n-1}\right] \quad \text{— (1)}$$

From Figure 2,

$$\alpha_n + \theta_{n+1} = \pi$$

$$\Rightarrow \alpha_n = \pi - \theta_{n+1}$$

From (1),

$$\alpha_n = \pi - \operatorname{Tan}^{-1}\left[x \cdot 2^n\right] \quad \text{— (2)}$$

$$\phi_n + \theta_n + \alpha_n = \pi \quad \text{— (3)}$$

$$\phi_n + \operatorname{Tan}^{-1}(x \cdot 2^{n-1}) + \pi - \operatorname{Tan}^{-1}(x \cdot 2^n) = \pi$$

$$\Rightarrow \phi_n = \operatorname{Tan}^{-1}(x \cdot 2^n) - \operatorname{Tan}^{-1}(x \cdot 2^{n-1}) \quad \text{— (4)}$$